

2. Methods

This section describes the numerical model in the order in which the code executes it: the porous medium is generated and conditioned, the flow field is solved on the resulting pore space, solute transport is solved on that flow field, and the reactions are evaluated at each transport step. Two- and three-dimensional simulations use the same code throughout except for the flow lattice, and the differences are noted where they arise.

The intention is that every choice made in the implementation can be justified when questioned, so each is stated with its reason rather than only its result. Section 2.1 defines three conventions that the rest of the description depends on.

2.1. Conventions Used Throughout

Three conventions are used in every section below. Each is stated here with the reason for it, because each affects how the reported numbers should be read.

2.1.1. *Normalised concentrations*

Every solute is divided by the concentration at which it enters the domain, so that a value of one denotes a solute as concentrated as the feed and zero denotes its absence. The reason is that only ratios of concentration appear in the transport and reaction equations once they are written in dimensionless form, so a single simulation applies at any feed concentration and need not be repeated when the feed is changed. Normalisation also makes the kinetic parameters directly interpretable: a half-saturation constant of 0.15 means the reaction proceeds at half its maximum rate once the solute has fallen to 15% of its feed value, whatever that value was in mol L^{-1} . Each solute is scaled by its own feed rather than by a common reference, which places all species on the same numerical range and moves the ratio of the two feed concentrations into the stoichiometric coefficient (Section 2.6).

2.1.2. *Interstitial velocity*

The velocity scale used in the Péclet and Damköhler numbers, written $\bar{u}$, is the mean of the velocity component along the flow direction, taken over all pore voxels. It is not the mean speed. Water in a tortuous pore space follows a winding path, so the mean speed exceeds the mean axial component; the ratio was 1.140 on one of the flow fields computed here and varies with pore structure.

The axial component is used for two reasons. It is the rate at which solute actually advances towards the outlet, which is the quantity a Péclet number defined on the sample length compares against diffusion. And it is equal to the Darcy flux divided by the porosity, q/ϕ , which is the interstitial velocity used in the conventional definition of the Péclet number, so that the values reported here are comparable with published ones. Using the mean speed would inflate the reported Péclet number by a tortuosity factor that differs between domains.

2.1.3. Flux formulation

The transport equation is discretised by finite volumes: for each voxel, the rate of change of its solute content is the net amount crossing its faces. This is written for one voxel i as

$$dC_i/dt = -(1/\Delta x) \sum_{\text{faces}} (J_{\text{adv}} + J_{\text{dis}}), \quad (1)$$

where the summation runs over the faces of the voxel and the two fluxes are advection and diffusion evaluated on the face rather than at the voxel centre,

$$J_{\text{adv}} = u_f C_f, \quad J_{\text{dis}} = -D_f (\partial C / \partial x)_f. \quad (2)$$

Equation (2) is Fick's first law together with advective transport; the only distinction from a cell-centred discretisation is where they are evaluated. That distinction has three consequences that matter here, and they are the reason for the choice. Mass is conserved exactly, because the flux one voxel sends across a face is by construction the flux its neighbour receives. The no-flux condition at a grain surface is imposed exactly, because a face adjoining solid simply carries zero flux, whereas a cell-centred scheme must approximate it. And a diffusion coefficient that varies in space, as it does between water and biofilm, can be evaluated correctly at the interface (Section 2.5.1).

Lattice units are used throughout unless stated otherwise: lengths in voxels, time in lattice steps. Parameters carrying metric units are identified explicitly, for example dx in m and D in $\text{m}^2 \text{s}^{-1}$.

2.2. Porous Media Generation

Computational domains were constructed as regular voxel grids, 148×64 voxels in two dimensions and $32 \times 32 \times 32$ to $64 \times 64 \times 64$ voxels in three dimensions, with flow imposed along the x-direction. Domains were generated at porosities spanning the range over which the pore space percolates in each dimension.

Gaussian porous media were used throughout, since they reproduce the bicontinuous pore distributions of geological materials such as soil aggregates and sandstone. A random number between 0 and 1 was first assigned to each voxel. A Gaussian filter with a standard deviation of

$\sigma = 3$ voxels was then applied to smooth the random field, and a threshold was set to obtain the desired porosity by classifying voxels above it as solid. The first and last four slices along the flow direction were forced open to provide the solvers with a clean inlet and outlet face.

The percolation threshold of a thresholded Gaussian field differs strongly between the two dimensions, occurring near a porosity of 0.59 in two dimensions and near 0.20 in three, because a path obstructed within a plane may be circumvented in a volume. Porosities were therefore sampled between 0.55 and 0.85 for two-dimensional domains and between 0.30 and 0.50 for three-dimensional domains. For the same reason, a two-dimensional domain is not a statistically equivalent representation of a three-dimensional one, and this is accounted for when two-dimensional data are used to supplement three-dimensional training (Jung et al. 2023).

Two alternative sources of geometry are supported. Domains may be read from the published two-dimensional dataset of Jung et al. (2023), or from files written in the material codes used by the reference simulator CompLaB, which allows the same structures to be run by both codes. In all cases the domain is stored as an integer array in which 0 denotes solid, 1 denotes the solid voxels adjacent to pore space, 2 denotes pore, and an optional fourth code denotes biofilm.

2.3. Domain Conditioning

Before any equation was solved, pore voxels that could not be reached from the inlet face through connected pore space were reclassified as solid. The retained cluster is the one connected to the inlet rather than the largest cluster in the domain; these are not always the same, and on one test domain the largest cluster contained 6936 voxels but did not intersect the inlet face, while the inlet-connected cluster contained 5736 voxels. This step was performed last, after every operation that can alter connectivity, since sealing the lateral faces afterwards can isolate regions that the connectivity test had already accepted.

Two distance fields were then computed on the conditioned domain and stored with each simulation. The geodesic distance from the inlet, ϕ , is the length of the shortest path through pore space and satisfies the eikonal equation

$$|\nabla\phi| = 1, \quad \phi = 0 \text{ on the inlet face}, \quad (3)$$

where the solid phase is treated as impassable. Equation (3) was solved by iterative relaxation over pore neighbours; fast marching (Sethian 1996) and fast sweeping (Zhao 2005) solve the same equation and may be substituted, subject to the caveat in Section 2.9. The Euclidean distance to the nearest solid voxel was computed by a standard distance transform and serves as a control: the two fields coincide in an open channel but differ by approximately a factor of two in

a tortuous pore space, where the straight-line distance implies access to regions that are in fact enclosed by solid.

Where attached biomass was represented explicitly, a biofilm layer of prescribed thickness was grown on the grain surfaces by binary dilation of the solid phase intersected with the pore space. Biofilm voxels are not solid: solutes diffuse through them at a separate diffusion coefficient (Section 2.5). The coated domain was used for transport only, while the flow field, the distance fields and the connectivity test were computed on the uncoated domain, since coating narrows pore throats and can disconnect the sample.

2.4. Fluid Flow

Pore-scale flow was simulated with the Lattice Boltzmann method, which recovers the Navier–Stokes equations in the low-Mach limit through a Chapman–Enskog expansion (He and Luo 1997; Chen and Doolen 1998). The method was chosen over a direct discretisation of the Navier–Stokes equations because the pore geometry is already a voxel grid and requires no mesh, and because the no-slip condition on an arbitrarily complicated grain surface is imposed by a local operation on each solid voxel rather than by fitting a boundary. At the pore velocities considered here the Reynolds number is much smaller than unity and the recovered solution is that of Stokes flow,

$$\mu \nabla^2 \mathbf{u} = \nabla p, \quad \nabla \cdot \mathbf{u} = 0, \quad \mathbf{u} = 0 \text{ on grain surfaces}, \quad (4)$$

where $\mathbf{u}$ is the pore velocity, p the pressure and μ the dynamic viscosity. Because Eq. (4) is linear, the flow field was solved once per domain and reused for every parameter combination applied to that domain.

We implemented fluid flow using a two-dimensional lattice with nine discrete velocity directions (D2Q9) and a three-dimensional lattice with nineteen directions (D3Q19), following the standard formulation (Qian et al. 1992; d'Humières et al. 2002):

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) + \Omega_i + F_i, \quad (5)$$

where f_i represents the particle distribution function along direction i at position $\mathbf{x}$ and time t , $\mathbf{e}_i$ is the lattice velocity, Ω_i is the collision operator and F_i is the forcing term. The equilibrium distribution function is defined as

$$f_i^{\text{eq}} = w_i \rho \left[1 + (\mathbf{e}_i \cdot \mathbf{u})/c_s^2 + (\mathbf{e}_i \cdot \mathbf{u})^2/(2c_s^4) - (\mathbf{u} \cdot \mathbf{u})/(2c_s^2) \right], \quad (6)$$

where w_i are the lattice weights (4/9, 1/9 and 1/36 for D2Q9; 1/3, 1/18 and 1/36 for D3Q19), $\rho = \sum_i f_i$ is the local fluid density and $c_s^2 = 1/3$ is the lattice speed of sound. Equation (6) is an

expansion in the Mach number, and simulations were constrained to remain below $Ma = 0.05$, above which the compressibility error is no longer negligible. The distribution functions were stored in double precision, since the non-equilibrium component that carries the viscous stress is of the order of the driving force itself.

2.4.1. Collision operator

A two-relaxation-time (TRT) collision operator was used in preference to the single-relaxation-time BGK operator (Bhatnagar et al. 1954). With a single relaxation time, the effective position of the no-slip wall produced by the bounce-back scheme depends on the fluid viscosity (Ginzburg and d'Humières 2003; Pan et al. 2006), so that the computed permeability of a fixed geometry varies with a purely numerical parameter. Each distribution function was therefore decomposed into symmetric and antisymmetric components about the opposite direction,

$$f_i^+ = (f_i + f_{\bar{i}})/2, \quad f_i^- = (f_i - f_{\bar{i}})/2, \quad (7)$$

where $\bar{i}$ denotes the direction opposite to i , and the two components were relaxed at separate rates:

$$\Omega_i = -\omega^+ (f_i^+ - f_i^{+e}) - \omega^- (f_i^- - f_i^{-e}). \quad (8)$$

The symmetric rate $\omega^+ = 1/\tau$ sets the kinematic viscosity, $\nu = c_s^2(\tau - 0.5)$ in lattice units, where τ is the dimensionless relaxation time. The antisymmetric rate ω^- is free and was fixed through the magic parameter

$$\Lambda = (1/\omega^+ - 1/2)(1/\omega^- - 1/2) = 3/16, \quad (9)$$

which places the bounce-back wall at the midpoint of the link between the last fluid and first solid node for every value of τ (Ginzburg et al. 2008). The operator reduces to BGK when $\omega^- = \omega^+$, which with Eq. (9) occurs at $\tau = 0.9330$. Measured wall positions, expressed as a fraction of the link where 0.5 denotes the midpoint, are given in Table 1.

τ	$H = 4$	$H = 8$	$H = 20$
0.55	0.437	0.470	0.489
0.80	0.467	0.492	0.495
0.933	0.499	0.501	0.500
1.20	0.598	0.550	0.520
2.00	1.100	0.834	0.637

Table 1. Effective wall position obtained with a single relaxation time, for channels H voxels wide. With the magic parameter of Eq. (7) the position is 0.500 at every relaxation time tested.

At $\tau = 2$ a four-voxel throat therefore transmitted 73% more flow than the analytical solution, and the discrepancy propagates directly into the computed permeability.

2.4.2. Driving force and boundary conditions

A pressure gradient that is uniform along the flow direction is equivalent to a uniform body force per unit volume, and the body force was used in preference to imposed inlet and outlet pressures because it requires no boundary treatment on those faces:

$$-\partial p / \partial x = g, \quad \text{applied as } F_i = w_i (1 - \omega^- / 2)(e_i \cdot g) / c_s^2. \quad (10)$$

where g is the imposed pressure gradient. Equation (10) follows the forcing scheme of Guo et al. (2002), with half of the force also added to the momentum when the velocity is recovered, $pu = \Sigma_i e_i f_i + (\Delta t / 2) F$. For a uniform body force the first-order forcing term is antisymmetric under $e_i \rightarrow -e_i$ and therefore carries the antisymmetric relaxation rate; using ω^+ in its place reduced the peak channel velocity by 18% at $\tau = 0.8$ and by 80% at $\tau = 0.55$, while leaving the profile visually parabolic.

Streaming was periodic along the flow direction, so that fluid leaving the outlet re-enters at the inlet. This avoids prescribing a velocity or pressure profile on those faces, which would impose a structure on the flow near them that the pore geometry did not produce. No-slip conditions at the solid–fluid interfaces were enforced using the halfway bounce-back scheme, in which the distribution functions arriving at a solid node are reflected into their opposite directions. The populations reversed are those that arrived rather than those produced by the current collision, which places the wall at the midpoint of the link; the effective aperture of a channel H voxels wide is therefore $H + 1$. No-flow boundaries were imposed along the lateral domain boundaries.

2.4.3. Setting the flow rate

Because Stokes flow is linear, the magnitude of the imposed gradient does not affect any dimensionless group. The velocity field was normalised after the solve so that its mean axial component over the pore space is unity, and the Péclet number was then imposed through the solute diffusion coefficient alone (Section 2.8). The magnitude of g was therefore chosen only to keep the simulation within the range of validity of Eq. (6): each domain was first solved at a small trial gradient, the peak velocity measured, and g rescaled so that $Ma \approx 0.02$. Applying one fixed gradient to every domain instead gave Mach numbers of 0.078 to 0.094.

The pressure gradient corresponding to a prescribed Péclet number follows from the same linearity and is reported with each simulation. Two domains of identical size and comparable

porosity required gradients differing by a factor of two for the same Péclet number, reflecting their different permeabilities.

The flow solution was iterated until the mean pore velocity was stationary,

$$| \langle |u| \rangle_n - \langle |u| \rangle_{n-m} | / \langle |u| \rangle_n \leq 10^{-6}, \quad (11)$$

evaluated every $m = 50$ iterations. The number of iterations required varied with both the relaxation time and the pore structure: on a 148×64 domain at a porosity of 0.63 it ranged from 4450 iterations at $\tau = 0.55$ to 19950 at $\tau = 2.00$. Terminating instead at a fixed iteration count underestimated the mean velocity by 9.8% in three dimensions, which propagates directly into the reported Péclet number.

2.5. Solute Transport

Each dissolved species obeys an advection–dispersion–reaction equation on the pore space,

$$\partial C_j / \partial t + \nabla \cdot (u C_j) = \nabla \cdot (D_j \nabla C_j) + R_j, \quad (12)$$

where C_j is the normalised concentration of species j , D_j its diffusion coefficient and R_j the net reaction rate from Section 2.6. A no-flux condition applies on every grain surface. No dispersion tensor is introduced: at the pore scale the spreading that a continuum model represents through dispersivity is resolved explicitly by the velocity field (Steefel et al. 2005; Molins et al. 2012).

Equation (12) was discretised on the same grid as the flow using the flux formulation of Section 2.1.3. The alternative, in which the voxel's own value replaces a neighbour that lies inside the solid, does not impose zero flux but removes one term from the balance. In a closed domain with a divergence-free recirculating flow this produced a mass drift of 1.35% with no obstacle present and 4.03% around a stepped grain surface, against 0.0003% and 0.0008% for the flux formulation.

2.5.1. Dispersive flux

The diffusive flux across the face separating voxels i and $i+1$ was evaluated as

$$F_{\text{dis}} = D^* (C_{i+1} - C_i) / \Delta x, \quad D^* = 2 D_i D_{i+1} / (D_i + D_{i+1}), \quad (13)$$

where D^* is the harmonic mean of the two cell diffusion coefficients. The harmonic mean is the series resistance of the two half-cells and is exact for a layered medium (Patankar 1980). It is required wherever the coefficient is not uniform, which here means at every pore–biofilm interface: using the local coefficient makes the flux discontinuous across such a face, and a

biofilm assigned one twentieth of the pore diffusion coefficient then produced an almost linear concentration profile of 0.310 where the flux-continuous solution gives 0.041.

2.5.2. Advective flux

First-order upwind differencing, although stable and monotone, is equivalent to solving Eq. (12) with an additional numerical diffusion coefficient of $|u|\Delta x/2$ per direction. In the units used here $\Delta x = 1$ and the normalised mean velocity is unity, so this spurious term is of order 0.5 while the physical coefficient is L/Pe ; the two are equal at a Péclet number of approximately $2L$. Since the purpose of these simulations is to label each domain by its Péclet number, the advective flux was instead reconstructed from the upwind side with a limited slope (van Leer 1979):

$$C_f = C_u + \frac{1}{2} \varphi(r)(C_d - C_u), \quad r = (C_u - C_{uu})/(C_d - C_u), \quad (14)$$

where C_u is the upwind cell, C_{uu} the next cell upwind, C_d the downwind cell and C_f the reconstructed face value, together with the minmod limiter

$$\varphi(r) = \max[0, \min(1, r)]. \quad (15)$$

The limiter vanishes at a local extremum, where the two differences have opposite signs, so the scheme reverts to upwind differencing there and the total variation cannot increase (Harten 1983; Sweby 1984). Minmod was preferred to the more common van Leer limiter because the latter is compressive: it does not merely omit numerical diffusion but removes physical dispersion, sharpening fronts that should be spreading (Roe 1986; LeVeque 2002). Table 2 compares the two against the analytical solution.

Péclet	analytical	first-order upwind	minmod	van Leer
20	36.7	36.6	34.3	34.3
50	23.2	26.1	22.5	22.3
100	16.4	20.9	16.0	15.6
300	9.5	15.8	9.0	7.7

Table 2. Width of a concentration front, measured between the 10% and 90% levels, after travelling half of a 64-voxel domain, compared with the complementary error function solution. Widths are in voxels.

The resolution limit of the grid may be expressed as a grid Péclet number, $Pe_g = u\Delta x/D = Pe/L$. Above $Pe_g \approx 2$ the concentration front is narrower than the voxels intended to resolve it and no scheme on that grid reproduces the imposed Péclet number; the value is reported for every simulation.

2.5.3. Divergence correction

Written in flux form, the advective term is identically $-u \cdot \nabla C - C(\nabla \cdot u)$, and the second term vanishes for an incompressible flow. A Lattice Boltzmann velocity field is only approximately solenoidal, however, and the residual acts as a source proportional to the local concentration. It does not cancel on average: in a $32 \times 24 \times 24$ domain at $Pe = 1.17$ the electron acceptor, which is only consumed, reached a concentration of 1.037 against an inlet value of 1.000. The term was therefore evaluated from the same face velocities used for the fluxes and subtracted, leaving $-u \cdot \nabla C$, which cannot generate a new maximum. On an exactly divergence-free field the correction is identically zero.

2.5.4. Boundary and initial conditions

A fixed concentration was imposed on the inlet face for each species that is supplied in the feed, a zero-gradient (Neumann) condition was applied at the outlet, and no-flux conditions were applied at the grain surfaces and the lateral boundaries. Species that are not supplied in the feed were left unconstrained at the inlet rather than fixed at zero: imposing a zero Dirichlet condition on the attached biomass suppressed the biotic reaction on precisely the face where both substrates are most abundant.

The initial concentration of every dissolved species was zero unless specified otherwise, representing a domain of clean water before the feed arrives. Attached biomass was present from the start, distributed uniformly through the pore space or, where the domain carries a biofilm phase, restricted to the biofilm voxels.

2.6. Reaction Network

Two reactions were implemented and may be enabled independently:



where Ac is the electron donor, A the electron acceptor, P the reaction product and Bio the microbial biomass. The stoichiometric coefficient α is expressed in the normalised concentrations and therefore equals the molar ratio multiplied by Ac_0/A_0 ; for equimolar consumption with inlet concentrations of 2.0 and 5.0 mmol L⁻¹ this gives $\alpha = 0.4$. Setting $\alpha = 1$ with equal inlet concentrations would make the donor and acceptor obey identical equations.

2.6.1. Biotic reaction

The microbially mediated rate follows dual Monod kinetics, saturating independently in the donor and the acceptor (Monod 1949; Rittmann and McCarty 2001):

$$R_{\text{bio}} = k \cdot [A_c/(K_{Ac} + A_c)] \cdot [A/(K_A + A)] \cdot (Bio/Bio_0), \quad (17)$$

where k is the rate constant, K_{Ac} and K_A are the half-saturation constants normalised by the corresponding inlet concentrations, and Bio_0 is the initial biomass. The individual species rates are $-R_{\text{bio}}$ for the donor, $-\alpha R_{\text{bio}}$ for the acceptor, $(1 - Y)R_{\text{bio}}$ for the product and $YR_{\text{bio}} - k_d \cdot Bio$ for the biomass, with Y the yield and k_d the first-order decay constant. The product receives $(1 - Y)$ rather than unity because all species are normalised against the donor inlet concentration, so that assigning the product a coefficient of unity would create $1 + Y$ moles of donor equivalent from one.

A half-saturation constant of zero denotes a first-order dependence on that species, with the constant absorbed into k , rather than Monod kinetics with a vanishing constant. The biomass factor is held at unity by default, since Bio_0 is already contained in the Damköhler number; a formulation in which the rate follows the instantaneous biomass is also available. Biomass was treated as attached to the grain surfaces and was therefore neither advected nor diffused, which is the appropriate representation of a biofilm; a planktonic formulation is available, and with it approximately three quarters of the biomass was lost by washout rather than by decay.

2.6.2. Abiotic reaction

The abiotic reaction is purely chemical and proceeds in the absence of microorganisms. It is first order in the species it consumes:

$$R_{\text{abio}} = k_s C_r \sigma(x), \quad (18)$$

where k_s is the surface rate constant, C_r is the reacting species and $\sigma(x)$ is a surface weighting. The reacting species is the donor when the biotic reaction is disabled and the product otherwise, since only the biotic reaction generates the product. The reaction converts the reactant to the product in a 1:1 ratio, or removes it entirely where a terminal loss such as precipitation is intended (Yoon et al. 2012).

Where the reaction is restricted to grain surfaces, as for a mineral-surface process, $\sigma(x) = n_f(x)/2n_d$ on pore voxels and zero elsewhere, with n_f the number of faces adjoining solid and n_d the number of spatial dimensions. A voxel within a narrow throat adjoins solid on several faces and presents correspondingly more reactive area per unit volume; on one test domain 363 surface voxels carried 494 solid-adjoining faces, and weighting them equally made the abiotic Damköhler number inconsistent between domains of differing specific surface area.

2.6.3. Operator splitting and rate limiting

Transport and reaction were advanced by operator splitting: the transport operator was applied first, and concentrations were then updated by the reaction rates evaluated from the local concentrations at that step. Splitting keeps the transport operator linear and allows the reaction to be evaluated independently in each voxel, so that a change to the reaction network requires no change to the transport scheme. Because the time step is derived from a linearised rate constant that is not constant when the rate follows the instantaneous biomass, the extent of reaction within one step was limited to half of the amount the scarcest reactant can supply, $R \leftarrow \min[R, \frac{1}{2} \min(Ac/\Delta t, A/\alpha\Delta t)]$. Where the reaction is slow this limit is never reached and the solution is unchanged; without it, the biomass-coupled formulation produced a product concentration of 505 against an inlet value of 1.

2.7. Time Integration and Convergence

Transport was integrated explicitly. An implicit scheme would remove the stability restriction on the time step, but the reaction terms are nonlinear and would require a nonlinear system to be solved over the whole pore space at every step; since the step is in any case limited by the reaction rate at the Damköhler numbers of interest, the restriction costs little. The step is bounded simultaneously by advection, dispersion and reaction,

$$\Delta t = 0.4 / [\sum_a \max|u| + 2 n_d D_{\max} + k_{lin} + k_s + k_d], \quad (19)$$

where the summation is over spatial directions, $D_{\max}$ is the largest diffusion coefficient among the species and k_{lin} is the linearised reaction rate. The three processes appear in a single limit rather than as separate limits of which the smaller is taken, since the latter is not the same quantity. The linearised rate is used because a Monod term $kC/(K + C)$ has slope k/K as C approaches zero, which exceeds k ; using k itself allowed the donor to be driven negative and clipped at zero while the product still received the full increment, giving outlet product concentrations of up to 5.093 against an inlet value of one.

The physical time step follows from the lattice diffusivity as $\Delta t_{phys} = \tilde{D} dx^2 / D$, where $\tilde{D}$ is the lattice diffusion coefficient, dx is the grid spacing in metres and D is the aqueous diffusion coefficient in $m^2 s^{-1}$.

Simulations were advanced until every species had reached a steady state. The criterion is applied to the distance remaining rather than to the change over the preceding interval, since a field approaching a steady state does so geometrically and a small increment does not imply proximity to the limit:

$$\delta_n = |\langle C_j \rangle_n - \langle C_j \rangle_{n-m}| / s_j, \quad \text{remaining} \approx \delta_n r / (1 - r), \quad r = \delta_n / \delta_{n-m}, \quad (20)$$

where s_j is the largest mean concentration that species j has attained, and the criterion is applied to the least converged species rather than to the donor alone. Judging convergence on the donor terminated simulations while the biomass was still 5.6% from its final value, because the donor is supplied at the inlet and equilibrates first while the biomass grows on a product made from it. With Eq. (20), a tolerance of 0.002 leaves the slowest species within 0.2% of its final value. Concentration fields were stored at logarithmically spaced times, since the pore space fills rapidly at first and asymptotically thereafter, and simulations that reached the iteration limit before converging were flagged as transient.

2.8. Dimensionless Parameters and Sampling

Each simulation is characterised by six dimensionless quantities. The Péclet and Damköhler numbers are defined as

$$Pe = \bar{u} L / D, \quad Da = k L / \bar{u}, \quad Da_a = k_s L / \bar{u}, \quad (21)$$

where $\bar{u}$ is the mean axial pore velocity, L is the reference length, and k and k_s are the biotic and abiotic rate constants. The remaining three are the two normalised half-saturation constants and the yield. The velocity scale is the mean of the axial component rather than the mean speed; the two differ by the tortuosity, measured at a ratio of 1.140 on one flow field, and the ratio varies with pore structure.

The reference length may be taken as the sample length or as the characteristic pore width; the two differ by a factor of 9.25 on a 148-voxel domain with a 16-voxel pore width and affect Pe and both Damköhler numbers in opposite directions, so the choice is recorded with each simulation. The advective and diffusive definitions of the Damköhler number differ by a factor of Pe ; the advective definition is used throughout.

Péclet and Damköhler numbers were sampled log-uniformly, since they span several orders of magnitude and the order of magnitude is the quantity of interest. The two Damköhler numbers were drawn independently so that the contributions of the two reactions remain separable.

2.9. Verification and Limitations

The flow and transport solvers were verified against solutions that do not derive from them (Table 3). Every simulation was additionally screened before its results were retained, and rejected if any species exceeded the concentration it could have been produced from, became negative, developed a checkerboard pattern along the flow direction, reached the outlet before

crossing the mid-plane, was a rescaled copy of another species, remained identically zero despite being supplied at the inlet, or carried a nonzero reaction rate for a species absent throughout the domain.

test	reference solution	agreement
Plane Poiseuille flow, 2D and 3D	$u_{\max} = gL^2/8\nu$, $L = H + 1$	peak within 1%, profile within 0.001% for $\tau = 0.55$ to 2.0
Wall position	midpoint of the link	within 0.1 voxel at every τ
Recovered viscosity	$\nu = c_s^2(\tau - 0.5)$	within 1%
2D and 3D channels of equal aperture	each other	agree to five decimal places
Plug flow with first-order decay	$\exp(-Da)$	0.610 against 0.607 at $Da = 0.5$
Front width	complementary error function	within 3% for $Pe = 20$ to 100
Mass in a closed domain	conservation	drift 0.0003%
Mass around a grain	conservation	drift 0.0008%
Donor plus product	inlet concentration	within 2% at $Da = 20, 100, 400$
Field with 10% artificial divergence	no solute created	maximum 1.00000

Table 3. Verification of the flow and transport solvers.

Three limitations follow from the model formulation. The pore structure is fixed: no voxel is converted between pore and solid, so precipitation that occludes a throat, and the resulting feedback on the flow field, are not represented. There is no aqueous speciation or chemical equilibrium; each species is transported as written and reacts only through Eqs. (17) and (18). Finally, the geodesic distance field stored with each simulation was computed by neighbour relaxation, and substituting a fast marching solver at the prediction stage returns a distance scale approximately 17% smaller on the same geometry, so the two must not be interchanged between training and prediction.

References

Two entries are to be supplied by the author: Jung et al. (2023), the two-dimensional predecessor of this work, and the reference simulator CompLaB.

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